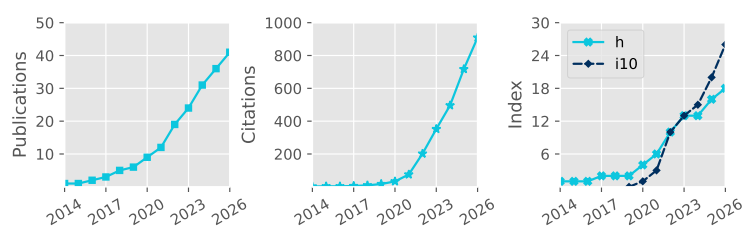


# Denis Sergeev

 Pronouns: he/him/his  
 University of Bristol, UK  
 denis.sergeev@bristol.ac.uk  
 0000-0001-8832-5288  
 dennisergeev.github.io  
 dennisergeev



Total Pub. **41**  
Refereed **39**  
First Author **8**  
Citations **909**  
h-index **18**

Updated: 14 Sept 2026

## Career history

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan 2025–now      | <b>Lecturer in Astrophysics</b><br>School of Physics, University of Bristol                                                                      |
| Sep 2021–Dec 2024 | <b>Postdoctoral Researcher</b><br>Project: Exascale Exoplanet Modelling<br>Department of Physics & Astronomy, University of Exeter               |
| Sep 2018–Aug 2021 | <b>Postdoctoral Researcher</b><br>Project: Climate Modelling of Rocky Exoplanets<br>Department of Mathematics & Statistics, University of Exeter |

## Academic Qualifications

|                   |                                                                                                                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oct 2014–Aug 2018 | <b>PhD in Meteorology</b><br>Thesis title (shortened): 'Characteristics of Polar Lows in the Nordic Seas' <a href="#">↗</a><br>School of Environmental Sciences, University of East Anglia<br>Supervisors: Ian A. Renfrew, Thomas Spengler, Stephen Dorling |
| Sep 2009–May 2014 | <b>Specialist Diploma (With Honours)</b><br>Thesis title: 'Idealised Numerical Modelling of Polar Mesocyclone Dynamics' <a href="#">↗</a><br>Department of Meteorology and Climatology, Moscow State University<br>Supervisor: Victor Stepanenko            |

## Funding and Awards

| Direct Funding, PI                       |                                                                               | Est. Total Value |
|------------------------------------------|-------------------------------------------------------------------------------|------------------|
| 2026                                     | Undergraduate Student Bursary   RAS                                           | £1500            |
| 2026                                     | Summer Student Placement Bursary   University of Bristol                      | ~£4500           |
| 2024                                     | Above & Beyond Silver Award   University of Exeter                            | £1000            |
| 2023                                     | Meeting Organisation Funding (Exoclimes VI and ExoSLAM)   RAS                 | £5000            |
| 2022                                     | Undergraduate Student Bursary (awarded; student declined)   RAS               | £1200            |
| 2017                                     | Best Presentation Award   CEEDA Symposium                                     | ~£100            |
| 2016                                     | Travel Bursary   Polar Prediction School                                      | ~£1000           |
| 2015                                     | Travel Award   High-Latitude Dynamics workshop                                | ~£1000           |
| 2014                                     | Lord Zuckerman PhD scholarship   School of Environmental Sciences, UEA        | ~£112 000        |
| 2014                                     | Young Scientist Travel Award   EGU General Assembly                           | ~£200            |
| 2014                                     | Russian Academy of Sciences Young Scientist Medal                             | ~£1000           |
| Direct Funding, co-I                     |                                                                               |                  |
| 2025                                     | Meeting Organisation Funding (UKExoM 2026)   RAS                              | £2500            |
| 2025                                     | Isambard 3 Allocation (30,000 node-hours; PI: N.J. Mayne)   UKRI (Isambard 3) | ▪                |
| 2024                                     | UKSA Studentships: Mars Exploration Science                                   | ~£100 000        |
| 2024                                     | Research Software Engineer Support   DiRAC HPC                                | ~£45 000         |
| Observational Facilities Resources, co-I |                                                                               |                  |
| 2023                                     | JWST Cycle 2, 49.21 hours (GO 3838, PI: J. Kirk)                              | ▪                |

## Publications

| #  | (preprints in grey)                                                                                                                                                                                                         | Citations |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 41 | Christie, D. A., Zamyatina, M., Hébrard, E., Evans-Soma, T. M., et al. (incl. <b>Sergeev, D. E.</b> ), 2026, Benchmarking two chemical networks used in general circulation models of hot Jupiters, MNRAS <a href="#">↗</a> | ▪         |

- 40 Stevenson, E. T., Mak, M. T., Wolf, E., **Sergeev, D. E.**, et al., 2026, ThousandWorlds: A benchmark for climate emulation of potentially habitable exoplanets, arXiv:2606.18338 [↗](#) ■
- 39 Nicholls, H., Krissansen-Totton, J., Lichtenberg, T., Schaefer, L., et al. (incl. **Sergeev, D.**), 2026, Coupled atmoSPHERE Interior model Intercomparison (CHILI). I. Evolutionary Modelling – Primordial Magma Oceans of Earth and Venus, arXiv:2606.24757 [↗](#) ■
- 38 Lichtenberg, T., Schaefer, L., Krissansen-Totton, J., Miguel, Y., et al. (incl. **Sergeev, D. E.**), 2026, Coupled atmoSPHERE Interior model Intercomparison (CHILI)—Protocol Version 1.0: A CUISINES Intercomparison Project of Magma Ocean Models, Planet. Sci. J. [↗](#) 7
- 37 Claringbold, A. B., Fisher, C. E., Kirk, J., Ahrer, E., et al. (incl. **Sergeev, D. E.**), 2026, BOWIE-ALIGN: Sub-solar C/O ratio and metallicity atmosphere of the misaligned hot Jupiter HAT-P-30 b, MNRAS [↗](#) 5
- 36 Ahrer, E., Fairman, C., Kirk, J., Wakeford, H. R., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: weak spectral features in KELT-7b's JWST NIRSpec/G395H transmission spectrum imply a high cloud deck or a low-metallicity atmosphere, MNRAS [↗](#) 7
- 35 Mak, M. T., **Sergeev, D. E.**, Mayne, N. J., Zamyatina, M., et al., 2025, The impact of different haze types on the atmospheres and observations of hot Jupiters: 3D simulations of HD 189733b, HD 209458b, and WASP-39b, MNRAS [↗](#) 5
- 34 **Sergeev, D. E.**, McDermott, J. W., Woods, L., Braam, M., et al., 2025, Lightning activity on a tidally locked terrestrial exoplanet in storm-resolving simulations for a range of surface pressures, MNRAS [↗](#) 1
- 33 Meech, A., Claringbold, A. B., Ahrer, E., Kirk, J., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: substellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy, MNRAS [↗](#) 14
- 32 Kirk, J., Ahrer, E., Claringbold, A. B., Zamyatina, M., et al. (incl. **Sergeev, D. E.**), 2025, BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b, MNRAS [↗](#) 31
- 31 Penzlin, A. B. T., Booth, R. A., Kirk, J., Owen, J. E., et al. (incl. **Sergeev, D. E.**), 2024, BOWIE-ALIGN: how formation and migration histories of giant planets impact atmospheric compositions, MNRAS [↗](#) 44
- 30 **Sergeev, D. E.**, Boutle, I. A., Lambert, F. H., Mayne, N. J., et al., 2024, The Impact of the Explicit Representation of Convection on the Climate of a Tidally Locked Planet in Global Stretched-mesh Simulations, ApJ [↗](#) 12
- 29 Natchiar, S. R. M., Webb, M. J., Lambert, F. H., Vallis, G. K., et al. (incl. **Sergeev, D. E.**), 2024, Reduction in the Tropical High Cloud Fraction in Response to an Indirect Weakening of the Hadley Cell, JAMES [↗](#) 4
- 28 Zamyatina, M., Christie, D. A., Hébrard, E., Mayne, N. J., et al. (incl. **Sergeev, D. E.**), 2024, Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example, MNRAS [↗](#) 24
- 27 Mak, M. T., **Sergeev, D. E.**, Mayne, N., Banks, N., et al., 2024, 3D simulations of TRAPPIST-1e with varying CO<sub>2</sub>, CH<sub>4</sub>, and haze profiles, MNRAS [↗](#) 12
- 26 Villanueva, G. L., Fauchez, T. J., Kofman, V., Alei, E., et al. (incl. **Sergeev, D. E.**), 2024, Modeling Atmospheric Lines by the Exoplanet Community (MALBEC) Version 1.0: A CUISINES Radiative Transfer Intercomparison Project, Planet. Sci. J. [↗](#) 13
- 25 Kirk, J., Ahrer, E., Penzlin, A. B. T., Owen, J. E., et al. (incl. **Sergeev, D. E.**), 2024, BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history, RAS Techniques and Instruments [↗](#) 25
- 24 Mak, M. T., Mayne, N. J., **Sergeev, D. E.**, Manners, J., et al., 2023, 3D Simulations of the Archean Earth Including Photochemical Haze Profiles, J. Geophys. Res.: Atmospheres [↗](#) 12
- 23 **Sergeev, D. E.**, Mayne, N. J., Bendall, T., Boutle, I. A., et al., 2023, Simulations of idealised 3D atmospheric flows on terrestrial planets using LFRic-Atmosphere, Geosci. Model Dev. [↗](#) 17
- 22 Cohen, M., Bolasina, M. A., **Sergeev, D. E.**, Palmer, P. I., et al., 2023, Traveling Planetary-scale Waves Cause Cloud Variability on Tidally Locked Aquaplanets, Planet. Sci. J. [↗](#) 10
- 21 Eager-Nash, J. K., Mayne, N. J., Nicholson, A. E., Prins, J. E., et al. (incl. **Sergeev, D. E.**), 2023, 3D Climate Simulations of the Archean Find That Methane has a Strong Cooling Effect at High Concentrations, J. Geophys. Res.: Atmospheres [↗](#) 9
- 20 McCulloch, D., **Sergeev, D. E.**, Mayne, N., Bate, M., et al., 2023, A modern-day Mars climate in the Met Office Unified Model: dry simulations, Geosci. Model Dev. [↗](#) 9

- 19 Braam, M., Palmer, P. I., Decin, L., Ridgway, R. J., et al. (incl. **Sergeev, D. E.**), 2022, Lightning-induced chemistry on tidally-locked Earth-like exoplanets, MNRAS [↗](#) 24
- 18 Christie, D. A., Lee, E. K. H., Innes, H., Noti, P. A., et al. (incl. **Sergeev, D. E.**), 2022, CAMEMBERT: A Mini-Neptunes General Circulation Model Intercomparison, Protocol Version 1.0.A CUISINES Model Intercomparison Project, Planet. Sci. J. [↗](#) 9
- 17 **Sergeev, D. E.**, Fauchez, T. J., Turbet, M., Boutle, I. A., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). II. Moist Cases-The Two Waterworlds, Planet. Sci. J. [↗](#) 80
- 16 Turbet, M., Fauchez, T. J., **Sergeev, D. E.**, Boutle, I. A., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). I. Dry Cases-The Fellowship of the GCMs, Planet. Sci. J. [↗](#) 66
- 15 Fauchez, T. J., Villanueva, G. L., **Sergeev, D. E.**, Turbet, M., et al., 2022, The TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI). III. Simulated Observables-the Return of the Spectrum, Planet. Sci. J. [↗](#) 56
- 14 **Sergeev, D. E.**, Lewis, N. T., Lambert, F. H., Mayne, N. J., et al., 2022, Bistability of the Atmospheric Circulation on TRAPPIST-1e, Planet. Sci. J. [↗](#) 34
- 13 Cohen, M., Bollasina, M. A., Palmer, P. I., **Sergeev, D. E.**, et al., 2022, Longitudinally Asymmetric Stratospheric Oscillation on a Tidally Locked Exoplanet, ApJ [↗](#) 17
- 12 Fauchez, T. J., Turbet, M., **Sergeev, D. E.**, Mayne, N. J., et al., 2021, TRAPPIST Habitable Atmosphere Intercomparison (THAI) Workshop Report, Planet. Sci. J. [↗](#) 39
- 11 Terpstra, A., Renfrew, I. A., & **Sergeev, D. E.**, 2021, Characteristics of Cold-Air Outbreak Events and Associated Polar Mesoscale Cyclogenesis over the North Atlantic Region, J. Cli. [↗](#) 31
- 10 Renfrew, I. A., Barrell, C., Elvidge, A. D., Brooke, J. K., et al. (incl. **Sergeev, D.**), 2021, An evaluation of surface meteorology and fluxes over the Iceland and Greenland Seas in ERA5 reanalysis: The impact of sea ice distribution, Q. J. R. Meteorol. Soc. [↗](#) 81
- 9 Eager-Nash, J. K., Reichelt, D. J., Mayne, N. J., Hugo Lambert, F., et al. (incl. **Sergeev, D. E.**), 2020, Implications of different stellar spectra for the climate of tidally locked Earth-like exoplanets, A&A [↗](#) 29
- 8 **Sergeev, D. E.**, Lambert, F. H., Mayne, N. J., Boutle, I. A., et al., 2020, Atmospheric Convection Plays a Key Role in the Climate of Tidally Locked Terrestrial Exoplanets: Insights from High-resolution Simulations, ApJ [↗](#) 67
- 7 Joshi, M. M., Elvidge, A. D., Wordsworth, R., & **Sergeev, D.**, 2020, Earth's Polar Night Boundary Layer as an Analog for Dark Side Inversions on Synchronously Rotating Terrestrial Exoplanets, The Astrophysical Journal Letters [↗](#) 20
- 6 Renfrew, I. A., Pickart, R. S., Våge, K., Moore, G. W. K., et al. (incl. **Sergeev, D.**), 2019, The Iceland Greenland Seas Project, BAMS [↗](#) 30
- 5 **Sergeev, D.**, Renfrew, I. A., & Spengler, T., 2018, Modification of Polar Low Development by Orography and Sea Ice, Mon. Wea. Rev. [↗](#) 19
- 4 Shestakova, A. A., Toropov, P. A., Stepanenko, V. M., **Sergeev, D. E.**, et al., 2018, Observations and modelling of downslope windstorm in Novorossiysk, Dyn. Atm. Ocean. [↗](#) 9
- 3 **Sergeev, D. E.**, Renfrew, I. A., Spengler, T., & Dorling, S. R., 2017, Structure of a shear-line polar low, Q. J. R. Meteorol. Soc. [↗](#) 22
- 2 Spengler, T., Renfrew, I. A., Terpstra, A., Tjernström, M., et al. (incl. **Sergeev, D.**), 2016, High-Latitude Dynamics of Atmosphere-Ice-Ocean Interactions, BAMS [↗](#) 6
- 1 Eliseev, A. V., & **Sergeev, D. E.**, 2014, Impact of subgrid-scale vegetation heterogeneity on the simulation of carbon-cycle characteristics, Izv. Atmos. Ocean. Phy. [↗](#) 9

## Conferences and Seminars

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### Invited Talks

- May 2025 General circulation models for exoplanet atmospheres  
JpGU-AGU Joint Meeting 2026 | Chiba, Japan
- Oct 2025 CUISINES - a framework for exoplanet model intercomparison projects  
Atmospheric and interior evolution of planetary magma oceans | Leiden, the Netherlands
- Jun 2025 Atmospheric dynamics on other planets [↗](#)  
Durham HPC Days | Durham, UK

- Feb 2025 Exoplanet climate modelling with LFRic  
University of East Anglia | Norwich, UK
- May 2024 3D simulations of exoplanet atmospheres with the next-generation Met Office model  
University of Leicester | Leicester, UK
- Apr 2024 Shall I compare thee to a distant world? Inter-planet and inter-model comparative studies  
EGU General Assembly | Vienna, Austria
- Jul 2023 Simulations of idealised 3D atmospheric flows on terrestrial planets using LFRic-Atmosphere  
NASA GISS Seminar | Online
- Mar 2023 First results of using LFRic for exoplanet climate modelling  
NIWA Seminar | Wellington, New Zealand
- Feb 2023 Atmospheric dynamics and chemistry on exoplanets  
UQ Astro Group Meeting | Brisbane, Australia
- Feb 2023 Atmospheric dynamics and chemistry on exoplanets [🔗](#)  
UniSQ Exoplanet Group Seminar | Brisbane, Australia
- Feb 2023 Atmospheric dynamics and chemistry on exoplanets  
UNSW AstroSeminar | Sydney, Australia
- Apr 2022 Dichotomy of the atmospheric circulation on TRAPPIST-1e [🔗](#)  
NASA GISS Seminar | Online
- Jan 2022 Dichotomy of the atmospheric circulation on TRAPPIST-1e  
NASA GSFC Extrasolar Planets Seminar | Online
- Nov 2021 TRAPPIST-1 Habitable Atmosphere Intercomparison (THAI)  
MPIA APEX Exocoffee | Online
- May 2021 Overcast on TRAPPIST-1e [🔗](#)  
RCC MSU Geophysical Seminar | Online
- Sep 2020 Simulations of convection over a range of atmospheric conditions on TRAPPIST-1e [🔗](#)  
THAI Workshop | Online
- Apr 2020 Atmospheric convection plays a key role in the climate of tidally locked exoplanets [🔗](#)  
University of Reading Meteorology Seminar | Online
- Apr 2020 Atmospheric convection plays a key role in the climate of tidally locked exoplanets [🔗](#)  
NASA GISS Seminar | Online

## Contributed Talks

- May 2026 Lightning climatology on TRAPPIST-1e  
ELSI workshop | Tokyo, Japan
- Sep 2023 Introducing GeoVista - Cartographic rendering and mesh analytics powered by PyVista (joint talk)  
Met Office Seminar | Exeter, UK
- Jul 2022 Bistability of the atmospheric circulation on TRAPPIST-1e  
Rocky Worlds II | Oxford, UK
- Apr 2022 Dichotomy of the atmospheric circulation on TRAPPIST-1e  
Exoplanet Modelling in the James Webb Era II: Terrestrial planets and sub-Neptunes | Online
- Nov 2020 Explicit convection on tidally locked rocky exoplanets simulated with the UM nesting suite [🔗](#)  
Unified Model users workshop | Online
- Aug 2019 Simulations of moist convection on tidally-locked rocky exoplanets [🔗](#)  
Exoclimates V | Oxford, UK
- Jun 2019 North Atlantic polar mesoscale cyclones in ERA5 and ERA-Interim reanalyses [🔗](#)  
IGP workshop | Norwich, UK
- Apr 2019 Atmospheric convection on tidally-locked Earth-like exoplanets  
UK Exoplanet Community Meeting | London, UK
- Jun 2018 Modification of Polar Low Development by Sea Ice and Svalbard Orography [🔗](#)  
POLAR2018 | Davos, Switzerland
- Oct 2017 The influence of Svalbard orography and sea ice on polar low development [🔗](#)  
18th Cyclone Workshop | Sainte-Adèle, Canada
- Apr 2017 Polar lows and how background environment can influence their development [🔗](#)  
Cambridge Earth Systems Science EnvEast Doctoral Alliance Symposium | Cambridge, UK

- May 2016 Structure of the shear-line polar low south of Svalbard  
NORPAN meeting | Tokyo, Japan
- Apr 2016 Structure of the shear-line polar low south of Svalbard [↗](#)  
13th European Polar Lows Working Group Workshop | Paris, France

## Poster Presentations

- Nov 2025 Lightning climatology on rocky exoplanets in a global storm-resolving model  
CTR Wilson Meeting on Atmospheric Electricity | Bath, UK
- Jun 2024 The impact of convection on the climate of a tidally locked planet in stretched-mesh simulations  
Exoplanets 5 | Leiden, Netherlands
- Apr 2024 The impact of convection on the climate of TRAPPIST-1e in global stretched-mesh simulations  
EGU General Assembly | Vienna, Austria
- Apr 2024 The impact of convection on the climate of a tidally locked planet in stretched-mesh simulations  
UK Exoplanet Community Meeting | Birmingham, UK
- Nov 2022 Dry Modern-Day Mars Climate in the Met Office Unified Model  
UK Solar System Planetary Atmospheres | London, UK
- Sep 2022 Bistability of the Atmospheric Circulation on TRAPPIST-1e  
UK Exoplanet Community Meeting | Edinburgh, UK
- Jul 2015 Structure and dynamics of a shear-line polar low during a cold-air outbreak over the Norwegian Sea  
Royal Meteorological Society Student Conference | Birmingham, UK
- Mar 2015 Structure and dynamics of a shear-line polar low during a cold-air outbreak over the Norwegian Sea  
Dynamics of Atmosphere-Ice-Ocean Interactions in the High Latitudes workshop | Rosendal, Norway
- May 2014 Numerical modelling of polar mesocyclones dynamics diagnosed by the energy budget  
EGU General Assembly | Vienna, Austria
- Apr 2013 Impact of subgrid-scale vegetation heterogeneity on the carbon cycle  
EGU General Assembly | Vienna, Austria
- Apr 2013 Numerical modelling of polar mesocyclones generation mechanisms  
EGU General Assembly | Vienna, Austria

## Supervision

(Projects with me as the lead supervisor are in **bold**. Students who continued their academic career are underlined.)

### PhD Supervision

- Sep 2025–Sep 2029 **Alex Corbett** (U. Bristol)  
**Project: Convection on Sub-Neptunes**  
Co-supervisors: B. Shipway, Z. Leinhardt
- Sep 2025–Sep 2029 Will Luscombe  
Project: Forecasting Martian dust storms  
Co-supervisors: N. J. Mayne, M. Bate, B. Drummond
- Sep 2021–Apr 2025 Mei Ting (Martha) Mak (U. Exeter)  
Project: Hazes in Planetary Atmospheres  
Co-supervisors: N. J. Mayne, J. Manners, E. Hébrard

### MSc Supervision

- Sep 2025–May 2026 **Freya Evans & Daisy Green**  
Project: **Atmospheric Dynamics on Ice Giants**
- Sep 2025–May 2026 **Catherine Kerr & Lily Odhuba**  
Project: **Lightning Storms on Earth-like Exoplanets**
- Jan 2023–May 2024 Tom Batchelor, Luke Benzing, & Alex McGinty  
Project: Mars Atmosphere Modelling  
Co-supervisors: M. Bate, N. J. Mayne, D. McCulloch
- Sep 2020–Sep 2022 Danny McCulloch (MSci by Research)  
Project: Climate Modelling of Modern-Day Mars

|                   |                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apr 2021–Sep 2022 | Co-supervisors: M. Bate, N. J. Mayne<br>Meghan Plumridge (MSci by Research)<br>Project: Climate Modelling of Early Mars<br>Co-supervisors: M. Bate, N. J. Mayne                 |
| Jan 2021–May 2022 | Jasper Chadwick & Esse Sellwood<br>Project: Ocean Heat Transport on Rocky Exoplanets<br>Co-supervisors: F. H. Lambert, J. Eager-Nash                                            |
| Jan 2021–May 2022 | Isabelle Browne & Oakley Young<br>Project: Greenhouse Effect on Early Mars<br>Co-supervisors: F. H. Lambert, N. J. Mayne, J. Eager-Nash                                         |
| Jan 2020–May 2021 | Toby Ferrison<br>Project: Titan Climate Modelling<br>Co-supervisor: F. H. Lambert                                                                                               |
| Oct 2018–May 2019 | Jake Eager-Nash & David Reichelt<br>Project: Implications of Stellar Type on the Climate of Tidally Locked Terrestrial Exoplanets<br>Co-supervisors: F. H. Lambert, N. J. Mayne |

## Undergraduate and Summer Internship Supervision

|              |                                                                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun–Aug 2026 | <b>Evie Cushing</b><br>Project: <b>Radiative Transfer Modelling for Ice Giants</b>                                                                          |
| Jul–Sep 2022 | Oakley Young<br>Project: Ekman Ocean Model<br>Co-supervisors: J. Eager-Nash, F. H. Lambert                                                                  |
| Jun–Sep 2022 | <b>James McDermott &amp; Lottie Woods</b><br>Project: <b>Simulations of Lightning Storms on Tidally Locked Rocky Exoplanets</b>                             |
| Jun–Aug 2021 | Oakley Young<br>Project: Climate Modelling of Archean Earth<br>Co-supervisors: J. Eager-Nash, N. J. Mayne                                                   |
| Jun–Aug 2021 | Joshua Parkin & Esse Sellwood<br>Project: The Impact of Host Star Spectrum on the Climate of Rocky Exoplanets<br>Co-supervisors: J. Eager-Nash, N. J. Mayne |
| Jun–Aug 2019 | Isobel Parry<br>Project: Water Cycle on Proxima Centauri b<br>Co-supervisor: F. H. Lambert                                                                  |

## PhD Examinations

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2025 Maria Di Paolo | University of East Anglia | Supervisors: Manoj Joshi, David Stevens

## Teaching and Mentoring

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|          |                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------|
| 2026–now | Environmental Physics<br>Lecturer   University of Bristol   ~40 students                                            |
| 2025–now | Practical Physics III: Research Skills and Group Project<br>Tutor   University of Bristol   2 groups of ~7 students |
| 2025–now | Research Project in Physics<br>Supervisor & assessor   University of Bristol   ~10 students                         |
| Jul 2024 | Algorithms For Exascale Summer School <a href="#">🔗</a><br>Invited lecturer   University of Exeter   ~20 students   |
| Feb 2024 | Physics of Climate Change<br>Workshop lead   University of Exeter   ~30 students                                    |
| Jul 2023 | Climatemark Academy<br>Mentor   Online   3 groups of ~5 students                                                    |
| Jul 2023 | International Sustainability Summer School<br>Lecturer   University of Exeter   ~10 students                        |

- Jun 2023 Exoclimes Summer School in Atmospheres and Modelling (ExoSLAM) [🔗](#)  
Lecturer | University of Exeter | ~50 students
- 2016–2018 Introduction to Python in Environmental Sciences [🔗](#)  
Course creator & lead | University of East Anglia | ~50 students
- 2015–2017 Modelling Environmental Processes; Meteorology; Numerical Skills  
Teaching assistant | University of East Anglia

## Impact and Outreach

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- 2026 Astronomy on Tap | The Lansdown at Clifton, Bristol [🔗](#)
- 2023 Interview by the University of Exeter about my research [🔗](#)
- 2023 Interview by UKRI/STFC [🔗](#)
- 2023 Expert Scientist at the British Science Festival Climate Exhibition [🔗](#)
- 2022 Press releases: University of Exeter [🔗](#), American University [🔗](#), & INSU CNRS [🔗](#)
- 2020–now 3D visualisations of exoplanet simulations:  
'Cloudy Skies of Distant Exoplanets' [🔗](#) | University of Exeter Images of Research 2023  
'A refined look at tidally locked exoplanets' [🔗](#) | DiRAC HPC Research Image Competition 2023  
'Exoplanetary Atmospheres' [🔗](#) | Exeter Science Centre, Science as Art Gallery 2020  
'Dusty exoplanet atmospheres' [🔗](#) | Nature Press Release  
'Virtual Reality Exploration of Exoplanets' [🔗](#) | 360 VR video (contributor)
- 2019 Science consulting on the 'Exoplanet Explorers' videogame
- 2015 Blogging:  
Disastrous Disaster Movies [🔗](#)  
Polar Lows: What Fuels Arctic Hurricanes? [🔗](#)  
Worldwide Weird Weather Words [🔗](#)

## Research Leadership and Organisation of Scientific Meetings

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- 2024–now Co-lead of Climates Using Interactive Suites of Intercomparisons Nested for Exoplanet Studies (CUISINES) [🔗](#)
- Jan 2027 Bridging atmospheres and interiors: Combined modelling and experimental approaches for sub-Neptune exoplanets (RAS Specialist Discussion Meeting) | London, UK
- Sep 2026 BUFFET-6: Building a Unified Framework For Exoplanet Treatments (Co-chair) [🔗](#) | Online
- Jun 2026 Idealised modelling with LFRic: training course (Co-Chair and Instructor) | Exeter, UK | ~50 attendees
- Mar 2026 UK Exoplanet Community Meeting (SOC) [🔗](#) | Bristol, UK
- Oct 2025 Atmospheric and interior evolution of planetary magma oceans (SOC) [🔗](#) | Leiden, the Netherlands
- Sep 2025 BUFFET-5 (Co-chair) [🔗](#) | Bordeaux, France
- Jul 2025 Exoclimes VII (SOC) [🔗](#) | Montreal, Canada
- Jun 2025 Idealised modelling with LFRic (Chair) | Exeter, UK | ~50 attendees
- Oct 2024 BUFFET-4: Building a Unified Framework For Exoplanet Treatments (Co-chair) [🔗](#) | Online
- Jun 2024 What's Cookin' Doc? A CUISINES meeting (Chair) | Leiden, the Netherlands | ~20 attendees
- Jun 2023 ExoSLAM Summer School (Co-chair) [🔗](#) | Exeter, UK | ~50 attendees
- Jun 2023 Exoclimes VI (LOC) [🔗](#) | Exeter, UK | ~200 attendees
- Mar 2023 Challenge of Science Leadership Short Course | Exeter, UK
- Aug 2022 Exoclimatology Theory Group Summer Retreat | Mawgan Porth, UK | 15 attendees

## Reviewing and Academic Service

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- Journals Nat. Astron., MNRAS, Planet. Sci. J., Geophys. Res. Lett., ApJ, Planet. Space Sci., Q. J. R. Meteorol. Soc.
- Funding STFC Consolidated Grant, STFC ERF, NASA XRP, STFC Small Award
- Observations James Webb Space Telescope General Observer Programs (Exoplanets & Disks, Cycles 3 & 4)
- Membership Royal Astronomical Society, Europlanet Society

## Technical Skills

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|----------------------------------------|---------------------------------------------------------------------------------------|
| Numerical models                       | LFRic, Unified Model, SOCRATES, LAGRANTO, Isca                                        |
| Programming languages                  | Python, FORTRAN, MATLAB, NCL                                                          |
| Python libraries (user)                | cartopy, cython, iris, matplotlib, numpy, pandas, pyvista, xarray                     |
| Python libraries (creator/contributor) | aeolus, cartopy, pyvista, geovista                                                    |
| Parallel computing                     | Dask, MPI, OpenMP                                                                     |
| Version control                        | Git, Subversion                                                                       |
| Document preparation                   | L <sup>A</sup> T <sub>E</sub> X, Quarto, Jupyter Notebooks, Markdown, HTML, CSS, reST |

## Vocational Training

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|          |                                                                                         |
|----------|-----------------------------------------------------------------------------------------|
| Sep 2023 | Belbin Training <a href="#">↗</a>                                                       |
| Mar 2023 | Challenge of Science Leadership <a href="#">↗</a>                                       |
| Dec 2022 | Interview Training                                                                      |
| Jul 2020 | Writing Workshop for Climate Scientists                                                 |
| Mar 2020 | ESA JWST Master Class <a href="#">↗</a>                                                 |
| Jul 2019 | ICTP Summer School on Convective Organization and Climate Sensitivity <a href="#">↗</a> |
| Apr 2018 | Fortran Modernisation Workshop <a href="#">↗</a>                                        |
| Jan 2018 | Helicopter Underwater Escape Training Course (CA-EBS) <a href="#">↗</a>                 |
| Dec 2017 | Sea Survival Course                                                                     |
| Jun 2017 | Weather Presenting                                                                      |
| Feb 2017 | Level 1 First Aid for Field Work Course                                                 |
| Jan 2017 | Raspberry Pi Basics                                                                     |
| Apr 2016 | WWRP/WCRP/Bolin Center Polar Prediction School                                          |
| Dec 2014 | UK Met Office Unified Model Training                                                    |

## Vocational Experience

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|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Apr–Jun 2018 | Data Technician<br>Processing of meteorological data collected in the IGP field campaign <a href="#">↗</a>   University of East Anglia |
| 2015–2018    | Founder and Leader<br>Python Users Group <a href="#">↗</a>   University of East Anglia                                                 |
| Feb–Mar 2018 | Member of the Meteorology Team<br>The Iceland-Greenland Seas Project (IGP) field campaign   Akureyri, Iceland                          |
| Mar 2015     | Rapporteur<br>Dynamics of Atmosphere-Ice-Ocean Interactions in the High-Latitudes <a href="#">↗</a>   Rosendal, Norway                 |
| Oct 2013     | Research Intern<br>Geophysical Institute   University of Bergen, Norway                                                                |
| Aug–Sep 2013 | Trainee Forecaster<br>Forecast and Briefing Service   Main Aviation Meteorological Centre, Vnukovo Airport                             |
| Jul 2012     | Research Intern<br>A.M. Obukhov Institute of Atmospheric Physics   Moscow, Russia                                                      |

## Unsuccessful Funding Applications

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|      |                                          |
|------|------------------------------------------|
| 2026 | GW4 Building Communities Generator Fund  |
| 2025 | STFC Small Award                         |
| 2025 | Leverhulme Research Leadership Award     |
| 2025 | ERC Starting Grant                       |
| 2025 | 2nd GPU Embedded CSE (eCSE) support call |
| 2024 | 1st GPU Embedded CSE (eCSE) support call |
| 2023 | STFC Ernest Rutherford Fellowship        |
| 2021 | Leverhulme Trust Early Career Fellowship |